# ROS2 for ATOSFleetManagement Cheat Sheet

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## 1) Source environment
```bash
source /opt/ros/humble/setup.bash
source /home/sepast/atos_ws/install/setup.bash
```

## 2) Build only needed packages
```bash
cd /home/sepast/atos_ws
./scripts/build_atosfleetmanagement.sh
```

## 3) Start ATOSFleetManagement
```bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True
```

## 4) List running nodes
```bash
ros2 node list
```

## 5) List topics
```bash
ros2 topic list
```

## 6) Inspect COT topic data
```bash
ros2 topic echo /atos/truck_objects/cot
```

## 7) Publish a test COT message
```bash
ros2 topic pub /atos/truck_objects/cot std_msgs/msg/String "data: 'id=truck_1;distance_m=1000;tcp_conne
```

## 8) Publish two trucks close enough to trigger speed limit
```bash
ros2 topic pub /atos/truck_objects/cot std_msgs/msg/String "data: 'id=truck_1;distance_m=1000;tcp_conne
ros2 topic pub /atos/truck_objects/cot std_msgs/msg/String "data: 'id=truck_2;distance_m=1300;tcp_conne
```

## 9) Watch speed commands from TruckObjectControl
```bash
ros2 topic echo /atos/truck_objects/speed_command
```

## 10) Quick behavior check
- Gap `< 400 m` => target speed `30 km/h`
- Gap `< 200 m` => target speed `0 km/h`

## 11) Open GUI
- URL: `http://localhost:3000`

## 12) Stop all launched nodes
- Press `Ctrl+C` in the launch terminal.